

Data pathology

Layer

Mechanism

Verified by

Non-random curve-suite missingness; cross-well heterogeneity

Representation layer (§4.1)

missing/outlier flags + sentinels;
within-well robust z-score;
multi-scale windows; curve-suite fingerprint; strat./spatial context

RQ3 (Fig. 6)
RQ5 (Fig. 8)

Long-tailed labels;
ambiguous lithology boundaries

Learning layer (§4.2)

effective-number class weight $\times$
boundary weight $\times$
confidence weight

RQ2 (Tab. 4)
RQ4 (Fig. 7)

Single-backbone bias;
rare-class signal diluted in 12-way objective

Posterior layer (§4.3)

LightGBM + XGBoost + CatBoost
geometric fusion; coarse-to-fine;
one-vs-rest tail experts

RQ2 (Tab. 5)
RQ4 (Fig. 7)

Asymmetric geological misclassification cost

Decision layer (§4.4)

class-prior logit adjustment (τ);
tail gates; Bayes-risk decoding over FORCE penalty matrix

RQ1 (Fig. 4–5)
RQ6 (Fig. 9)